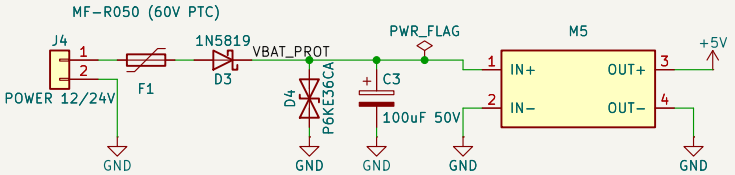
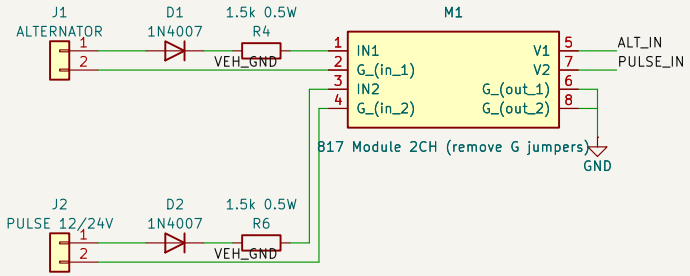
POWER: 12/24V -> F1 PTC -> D3 reverse diode -> D4 TVS + C3 -> M5 buck (adjust to 5.0V BEFORE connecting ESP32)



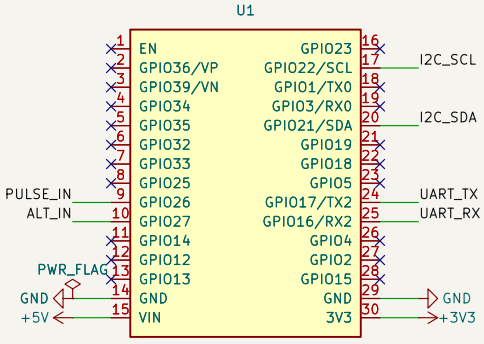
LM2596 buck module (set 5.0V)

I2C bus (400 kHz, module pull-ups)
DS3231 0x68 | AT24C256 0x50 (A0-A2 = GND, WP = GND)
MPU9250 AD0 = 3V3 -> 0x69 (0x68 would clash with RTC)
ZS-042 RTC board: check its battery-charge resistor (CR2032)
ESP32 VIN = +5V from M5 (buck module)
USB + VIN together only if DevKit has a VIN diode

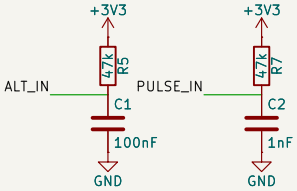
Hour Meter ESP32 — module wiring



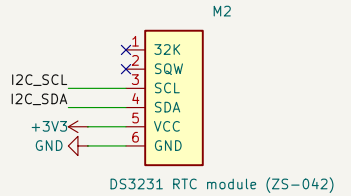
VEHICLE SIDE (isolated from ESP32)
J1 = alternator D+/L, J2 = 12/24V pulse; pin 2 = vehicle GND
D1/D2 = reverse-polarity protection
R4/R6 = keep 24V systems inside the module's 24V rating
817 MODULE: REMOVE BOTH G JUMPERS (keeps isolation)
Engine ON -> V1 pulled to G -> GPIO27 LOW (altActiveHigh = false)



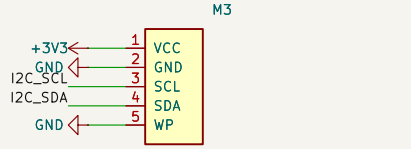
ESP32 DevKit V1 (30-pin)



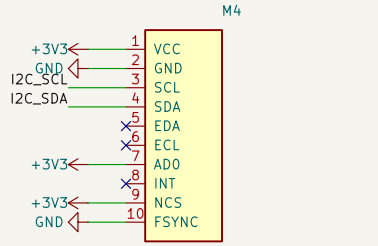
R5/R7 = 47k pull-ups (not 10k: module has ~3k in series with V)
C1 100nF = ignition-noise filter, C2 1nF = keeps fast pulses



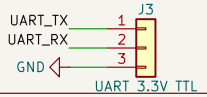
DS3231 RTC module (ZS-042)



AT24C256 EEPROM module (A0-A2 = GND, 0x50)



MPU9250 module (AD0 = 3V3, 0x69)



UART 3.3V TTL

Sheet: /
File: hour-meter-esp32.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 1/1